

Grouped Correlational Generative Adversarial Networks for Discrete Electronic Health Records

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Abstract—Using Generative Adversarial Networks (GANs) to generate synthetic Electronic Health Records (EHR) has attracted increasing attention. However, in existing approaches, the events in EHRs are treated as separate variables which are indiscriminately entered into the model, without taking into account the meaning and grouping of them. Besides, the efficacy of treatment is often neglected. In this paper, we first embed the efficacy information into the disease diagnosis, and then propose Grouped Correlational GAN (GcGAN) to explicitly learn inherent correlations between different groups of variables. We also introduce a dense connection to strengthen the generator capacity in GcGAN. Experimental results on real-world data demonstrate that the generated data from GcGAN are able to simulate real-world data in terms of distribution statistics. The results on multi-label treatment recommendation tasks show that GcGAN can boost the performances by augmenting the training dataset with the generated data and outperforms state-of-the-art approaches. It can also automatically distinguish between disease-specific drugs and adjuvant drugs, which enhances the model interpretability.

Index Terms—generative adversarial network, electronic health record, multi-label classification, deep learning.

I. INTRODUCTION

Electronic health records (EHRs) have been playing an essential role in medical information systems. However, access to EHR is limited due to the protection of patients' privacy in the data [1], [16], [23]. Hence the research of how to obtain a large amount of medical data has attracted increasing attention in medical& health research and applications.

In this work, our goal is to generate high-quality synthetic EHR data. Recently, Generative adversarial networks (GANs) [3] has shown amazing ability in generating data [11], [15], [29], [30]. Although most of them are used in image data, some studies have tried to apply GANs to

generate medical data [23]–[25]. However, EHR contains a large number of discrete features, which poses a challenge to exploiting applications of GANs. Further, existing research on applying GANs to the generation of medical data [7], [23]–[25] fails to fully consider features of EHR data. They treat all the variables (namely, diseases and drugs) equally and do not explicitly take into account the available meaning and grouping of the variables. In addition, model interpretability is important in medical applications, which is often neglected in GANs. Specifically, the existing GAN-based techniques fail to take into account the following characteristics of EHR data.

- Existing models usually implicitly build the relations between medical conditions and drugs for a single disease. However, the relations become more complicated when a patient has multiple diseases. In this case, various combinations of medical therapy have been prescribed in the treatment, as shown in Fig. 1. It requires the investigation of the relations between two sets of variables. How to effectively discover the relations between multiple diseases and treatment challenges data generation.
- Efficacy is often neglected in existing approaches. A treatment for a patient's disease will produce a curative effect, which characterizes the effectiveness of the treatment on different conditions, as shown in Fig. 1. Generally, including efficacy in the inpatient medical record is necessary in a real-world EHR database in a pediatric department of a hospital. A model is expected to learn the complex relationships between multiple diseases, treatments and efficacy.
- Most treatments include not only disease-specific drugs, but also adjuvant drugs and solvents that are added to enhance or extend primary therapy's effect, such as glucose, amino acids and saline solution. They have different distributions in the records. However, existing models treat the drugs equally. They cannot distinguish between them either.

To address the above issues, we first propose a new framework of *Grouped Correlational Neural Network* (Grouped CorrNN), which can learn the relationships between multiple

This work is supported by the Natural Science Foundation of China under Grant NO.61673324, NO.61772442 and NO.61836005, the Natural Science Foundation of Fujian under Grant NO.2018J01097, the Open Fund of Engineering Research Center for Medical Data Mining and Application of Fujian Province (MDM2018002) and National University Students Innovation Training Project of Xiamen University (2018X0224). Corresponding to: yang@xmu.edu.cn and y.zeng@tees.ac.uk

978-1-7281-1867-3/19/\$31.00 © 2019 IEEE

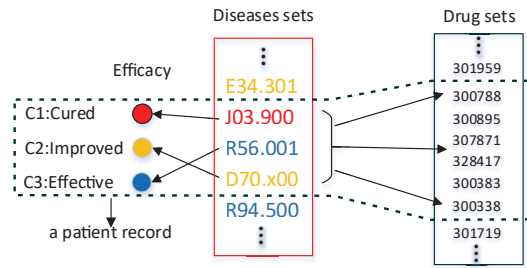


Fig. 1. The data in a patient record. C1, C2, and C3 indicate the three grades of efficacies. A typical patient record contains multiple diseases (here denoted as ICD-9 codes) and drugs (here denoted as the hospital's own codes). After treatments, the efficacies on the diseases can be observed.

diseases, treatments and corresponding efficacy by grouping the variables to different categories and explicitly considering their correlations. Then we develop an enhanced learning framework named GcGAN that combines Grouped CorrNN and T-GANs [26], and improve the model ability to produce high quality EHR data. In addition, we introduce the idea of Dense connection in the generator so that GcGAN can consider low-order features to accelerate the generator convergence and to better generate high-quality data. Our main contributions are listed below.

- The curative effect is embedded into a disease vector. We propose Grouped CorrNN to explicitly structure the correlation between diseases, drugs and efficacy.
- Grouped CorrNN endows the model with the ability of distinguishing the features among diseases, disease-specific drugs and adjuvant drugs, which can be respectively decoded from the grouped hidden layers. This property improves the model interpretability.
- We introduce dense connection for the first time in a GAN generator to improve model convergence, and further employ T-wGANs to enhance the stability of the model training.
- We conduct various experiments on a real-world EHR dataset. Besides the distribution statistics of the generated data, the results of multi-label treatment recommendation tasks also demonstrate that GcGAN outperforms state-of-the-art approaches.

II. RELATED WORK

Although GAN has demonstrated its power in various applications such as image-to-image translation [2], [17], text-to-image synthesis [9], [12], and video generation [8], its applications to EHR remain rare. RCGAN [24] is capable of generating real-values time series, while [4] is to generate continuous time series that display effects of exposure changes with a simple GAN architecture. Both of Them are used to generate continuous EHR data. EHR contains a large number of discrete features, which poses a challenge to exploiting applications of GANs. One way to generate discrete variables with GANs is to use reinforcement learning [6], namely SeqGAN [11]. However, SeqGAN generates word sequences by sampling one word at each time step, while our goal is

to generate multi-label binary variables with no sequential relationship between the labels, which hinders from using SeqGAN in EHR.

The state-of-the-art methods for generating discrete EHR data are ehrGAN [25] and medGAN [1]. The ehrGAN deals with high dimensional temporal EHR data, and segments the time dimension into disjoint 90-day windows and combines all the observations within each window. To generate samples with different lengths, it makes a special embedding mark at the end of each training record. Because it is temporal EHR data, the ICD-9 codes of diagnosis and medications are mapped to the embedding by using Word2vec [5], and then ehrGAN learns the embedding directly. After getting the generated data, it can map the embedding into EHR record by finding the nearest-neighbor with cosine distance for each feature. The ehrGAN is capable of boosting the classification performance with generated EHR data; however, they are mainly designed to generate EHR data for a single disease, such as heart failure and diabetes. The ability of generating multiple disease and drugs in one record is not guaranteed, which will be illustrated in our experiments.

Based on inputs of real-world patient records, medGAN can generate high-dimensional discrete variables (e.g., binary and count features) by using continuous values to approximate discrete values via a simple combination of an autoencoder and GANs. Although it is simple and efficient, it treats all the variable equally without considering the available information of the variables as well as the efficacy of the treatment.

The model interpretability is also one of the important criteria for evaluating the model, especially in the medical domain. An interpretable deep network RETAIN [27] utilizes a two-level network of attention to identify influential past visits and significant clinical variables within those visits. In GcGAN, by considering the meaning and grouping of variables, it can automatically distinguishes between different groups of variables. The reason of distinguishing between adjuvant drugs and disease-specific drugs in a record is two-folds. First, if the models can automatically distinguish the drugs, it will enhance the interpretability of the model and directly affect doctors' trust in the models. Second, incorporating the grouped correlation into GANs does help boost model performance, compared with just utilizing correlation information between disease and drugs.

III. GAN-BASED METHODS

A. The EHR structure & Efficacy embedding

Consider the EHR data $S = \{s_i\}_{i=1}^N$ with two views: binary disease variable X and binary drug variable Y . Each record s_i can be represented as a concatenation of these two views: $s_i = (x_i, y_i)$ where $x_i \in R^{d1}$ and $y_i \in R^{d2}$, and the j^{th} dimension indicates the absence or occurrence of the j^{th} variable in the patient record. $d1$ and $d2$ stand for the total number of diseases and total number of drugs in the respectively dataset.

Efficacy embedding: In order to effectively discover the relationship between drugs and diseases, we take into account the curative effect of treatments. In our EHR data, the efficacy

is divided into three grades, namely “Cured”, “Improved”, and “Effective”, denoted as C1, C2 and C3 respectively. Each disease has corresponding efficacy information. Subsequently $x_i \in R^{d1}$ is transformed into $x_i \in R^{3*d1}$ after the efficacy is embedded into the disease vector. The distribution of efficacy in the data is shown in Fig. 3(a).

B. Preliminary: GANs & CorrNN

GANs: The main idea of GANs is to train a generator $G(z)$ so that the generator produces the same distribution $q(x)$ as the real distribution $p(x)$, as shown in Eq.1, where $q(z)$ is a prior distribution, usually represented by a Gaussian distribution.

$$q(x) = \int \delta(x - G(z))q(z)dz \quad (1)$$

D and G play the following two-player minimax game with value function $V(G, D)$ to minimize the distribution distance of $p(x)$ and $q(x)$:

$$\min_G \max_D V(D, G) = \mathbb{E}_{x \sim p_{data}(x)} [\log D(x)] + \mathbb{E}_{z \sim q_z(z)} [\log(1 - D(G(z)))] \quad (2)$$

wGAN: wGAN [14] optimizes the following Wasserstein distance between $p(x)$ and $q(x)$,

$$\begin{aligned} W(p(x), q(x)) &= \inf_{\gamma \in \Pi(p(x), q(x))} \int_{x,y} \gamma(x, y) \cdot \|x - y\| d\gamma(x, y) \\ &= \inf_{\gamma \in \Pi(p(x), q(x))} \mathbb{E}_{(x,y)} \|x - y\| \end{aligned} \quad (3)$$

where $\gamma \in \Pi(p(x), q(x))$ represents a joint distribution of edge distributions $p(x), q(x)$. By using K-R dual [7], it can transform the problem into a solution to the optimal function:

$$W(p(x), q(x)) = \sup_{\|D\|_L \leq 1} \mathbb{E}_{x \sim p(x)} [D(x)] - \mathbb{E}_{x \sim q(x)} [D(x)] \quad (4)$$

$$\|D\|_L = \max_{x \neq y} \frac{|D(x) - D(y)|}{\|x - y\|} \quad (5)$$

where $\|D\|_L$ is the *Lipschitz* norm of the function $D(x)$, and $D(x)$ is the discriminator.

T-GANs: Recently, T-GANs is proposed to make training of GANs more stable [26]. It is a generalization of the relative discriminator based on RSGAN [13], which allows the introduction of real sample information into the process of training the generator to accelerate the entire training process. T-GANs optimize the distance between $p(x_r)q(x_f)$ and $p(x_f)q(x_r)$, with x_r, x_f representing the real sample and the generated sample, respectively. In this work, we use T-wGAN, which is the application of wGAN on T-GAN, so we use $T(x)$ to represent the discriminator. T-wGAN minimizes the *Wasserstein* distance of $p(x_r)q(x_f)$ and $p(x_f)q(x_r)$, which can be rewritten as:

$$\begin{aligned} &W(P(x_r, x_f), Q(x_r, x_f)) \\ &= \sup_{\|T\|_L \leq 1} \mathbb{E}_{(x_r, x_f) \sim P(x_r, x_f)} [T(x_r, x_f)] - \mathbb{E}_{(x_r, x_f) \sim Q(x_r, x_f)} [T(x_r, x_f)] \\ &= \sup_{\|T\|_L \leq 1} \mathbb{E}_{(x_r, x_f) \sim P(x_r, x_f)} [T(x_r, x_f) - T(x_f, x_r)] \end{aligned} \quad (6)$$

where P and Q are two joint distributions, defined as $P(x_r, x_f) = p(x_r)q(x_f)$ and $Q(x_r, x_f) = p(x_f)q(x_r)$. (x_r, x_f) can be regarded as an integrated variable, and $T(x)$ is the discriminator. Therefore, the min-max game during T-wGAN training can be defined as:

$$\arg \min_G \arg \max_{T, \|T\|_L \leq 1} \mathbb{E}_{(x_r, x_f) \sim p(x_r)q(x_f)} [T(x_r, x_f) - T(x_f, x_r)] \quad (7)$$

CorrNN: Correlational Neural Network [21] learns a common representation by minimizing the error of reconstructing the two views and takes into consideration the correlation in the two views while regenerating the input. It has been employed in adversarial training based correlated discrete data generation (CorrGAN) [28]. In this work, we employ it to enhance the correlation between diseases and treatment. The loss function of the CorrNN network is as follows:

$$\begin{aligned} L_{CorrNN} &= \sum_{i=1}^N L(D_{ec}(E_1(x_i)), s_i) + L(D_{ec}(E_1(y_i)), s_i) + \\ &L(D_{ec}(E_1(x_i) + E_2(y_i)), s_i) - \lambda corr(E_1(X), E_2(Y)) \end{aligned} \quad (8)$$

$$\begin{aligned} corr(E_1(X), E_2(Y)) &= \\ &\frac{\sum_{i=1}^N (E_1(x_i) - \bar{E}_1(x)) (E_2(y_i) - \bar{E}_2(y))}{\sqrt{\sum_{i=1}^N (E_1(x_i) - \bar{E}_1(x))^2 \sum_{i=1}^N (E_2(y_i) - \bar{E}_2(y))^2}} \end{aligned} \quad (9)$$

where λ is a coefficient that weighs reconstruction loss and correlation.

C. Our Approaches: GcGAN & Grouped CorrNN

The framework of GcGAN is shown in Fig. 2. We first propose Grouped CorrNN in the generation network, and then employ T-wGAN in adversarial training. The structure of Grouped CorrNN is displayed in Fig. 2(b). We structure the hidden layers in the framework and provide a new loss function accordingly. Similar to CorrNN, the encoders E1 and E2 correspond to the disease vector x_i and the drug vector y_i respectively. The difference is that Grouped CorrNN divides the representation vector of the hidden layer into different parts, corresponding to different groups of variables to be decoded. Considering that the disease-specific drugs and adjuvant drugs have different distributions, we divide the representation h_i into three parts, namely the disease representation h_{i1} , disease-specific drugs representation h_{i2} and adjuvant drugs representation h_{i3} . h_{i1} is decoded by D1 to obtain the disease vector x_i , and h_{i2} and h_{i3} are decoded by D2 to obtain disease-specific drugs $m1$ and adjuvant drugs $m2$ in the prescription respectively, where $m1 \cup m2 = y_i$ and

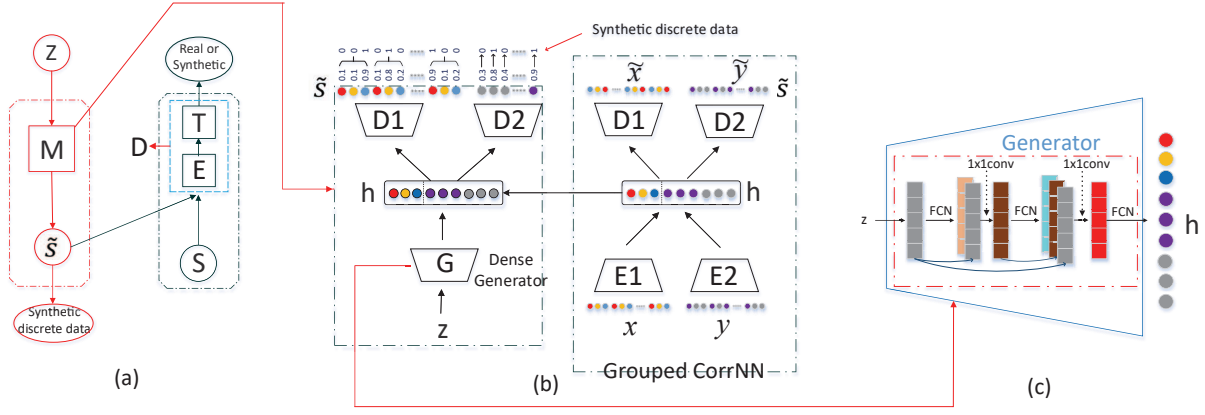


Fig. 2. The framework of GcGAN . (a) The general structure of GcGAN model. M are the generation network. D are the discriminative network, which employs T-GANs. It consists of an encoder E and a discriminator T. S represents real data. $\tilde{S}$ represents generated data. (b) The structure of M. G is the dense generator. E1 and E2 are encoders of Grouped CorrNN, and D1 and D2 are decoders. Z represents the noise vector, x represents the disease vector embedded with efficacies, y indicates the prescription in the treatment, and h represents the learned hidden layer representation. (c) The structure of G. FCN indicates a Fully Connected Neural Network.

$m1 \cap m2 = \emptyset$. To ensure that h_{i2} represents disease-specific drugs, h_{i2} should be able to decode disease x_i through D1.

Compared to the original CorrNN, our model not only more effectively finds the correlation between diseases and drugs, but also distinguishes between disease-specific drugs and adjuvant drugs. The additional loss is as follows:

$$L_h = \sum_{i=1}^N \lambda_1 (L_1(D_1(h_{i1}; 0; 0), x_i) + L_2(D_1(0; h_{i2}; 0), x_i)) + \lambda_2 (L_3(D_2(0; h_{i2}; h_{i3}), y_i) + L_4(D_2(0; h_{i2}; 0) \cup D_2(0; 0; h_{i3}), y_i) + L_5(D_2(0; h_{i2}; 0) \cap D_2(0; 0; h_{i3}), 0)) \quad (10)$$

where λ_1 and λ_2 represent the coefficients of the loss of the reconstructed disease vector x and the loss of the reconstructed drug vector y , respectively. In summary, the loss function of Grouped CorrNN is as follows:

$$L = L_{CorrNN} + L_h \quad (11)$$

As shown in Fig. 2(a), the generator G and Grouped CorrNN are integrated, and the parameters are trained simultaneously. Instead of learning the real EHR sample directly, G first learns the hidden vector h , and then synthesizes the real sample through the decoders D1 and D2. This not only alleviates the problem that the gradient cannot be effectively returned, but also distinguishes disease-specific drugs and adjuvant drugs in the prescription. Note that the outputs of decoders are continuous values. Hence we use the same strategy as medGAN to transform them to binary data. Specifically, for the drugs, if the value of a variable is greater than 0.5, we set it to 1, else set to 0. For the disease, because each disease is represented as a three dimensional vector with each dimension denoting a efficacy grade, we only set the component that has the maximum value and is greater than 0.5 to 1.

D. Enhanced Generator & Implementation details

To enhance the generator G, we resort to dense connection [19] in our model, as shown in Fig.2(c). To maximize

Algorithm 1 The optimization procedure of GcGAN

```

1: for n iterations until convergence do
2:   for k steps do
3:     Sample minibatch of m samples  $\{z^1, z^2, \dots, z^m\}$ 
       from noise prior  $p(z)$ 
4:     Sample minibatch of m examples  $\{s^1, s^2, \dots, s^m\}$ 
       from training data
5:      $s_r^i, s_f^i \leftarrow E(s^i), E(D_{ec}(G(z^i)))$ 
6:      $(s_r^i, s_f^i), (s_f^i, s_r^i) \leftarrow (s_r^i - s_f^i), (s_f^i - s_r^i)$ 
7:     Update discriminator T

        $\theta_T \leftarrow Adam(\nabla_{\theta_T} \frac{1}{m} \sum_{i=0}^m [T(s_f^i, s_r^i) - T(s_r^i, s_f^i)])$ 

8:   end for
9:   Sample minibatch of m samples  $\{z^1, z^2, \dots, z^m\}$  from
       noise prior  $p(z)$ 
10:  Sample minibatch of m examples  $\{s^1, s^2, \dots, s^m\}$  from
       training data
11:   $s_r^i, s_f^i \leftarrow E(s^i), E(D_{ec}(G(z^i)))$ 
12:   $(s_r^i, s_f^i), (s_f^i, s_r^i) \leftarrow (s_r^i - s_f^i), (s_f^i - s_r^i)$ 
13:  Update Grouped CorrNN and Generator G
        $\theta_L \leftarrow Adam(\nabla_{\theta_L} (L_{CorrNN} + L_h))$ 

        $\theta_G \leftarrow Adam(\nabla_{\theta_G} \frac{1}{m} \sum_{i=0}^m [T(s_r^i, s_f^i) - T(s_f^i, s_r^i)])$ 

14: end for

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the information flow between all layers in the network, dense connection connects the features of all the layers in front of the current layer. It is first used to generate discrete data in this work. For discrete EHR data we use the 1×1 convolution [10] to integrate the information flow from the previous layer and input it to the next layer. We train the GcGAN model by optimizing the generator and discriminator iteratively with *Adaptive Moment Estimation* (Adam) and the learning rate is set to 0.0002, which is depicted in Algorithm 1.

Here m , k are set to 64, 1 respectively. In addition, we apply spectral normalization (SN) [20] on discriminator to make the network satisfy *Lipschitz* continuity and use the batch normalization (BN) [22] on the generator to speed up the learning ability. In Grouped CorrNN, λ , λ_1 and λ_2 are empirically set to 1, 5, 2 respectively. We implement GcGAN with Tensorflow in a PC with *Intel i7*, *16GB RAM* and two *GTx 1080ti*.

IV. EXPERIMENTAL RESULTS

A. Data Description

The EHR data came from a real-world EHR database of about 20,000 unique medical events, between the year 2015 to 2017 from the pediatric department of a hospital. This database records complete medical information for pediatric inpatients during hospitalization. To protect patient privacy, all personal information was removed. After data cleaning, we finally got 17,000 unique records each of which consists of multiple diseases, treatments and corresponding efficacy. Note that including efficacy in the inpatient medical record is necessary, and how to define “Improved”, “Cured” and “Effective” is based on a uniform standard with high credibility and objectivity.

The dataset contains a total of 777 diseases and 748 drugs. The distribution of the efficacy, the number of disease in each record, and the number of drug data in each record, are shown in Fig. 3. The average number of the diseases per record is 3.1, and the average number of the drugs per record is 13.9. We use a set of diseases related ICD-9 codes to indicate medical conditions as well as drug prescriptions. After efficacy embedding, each record is transformed into a single fixed-size vector (the size is $777 * 3 + 748$) with binary features.

B. Models for comparison

We compare the performances of GcGAN and several state-of-the-art approaches on generating EHR data: 1)wGAN: Data generation is performed by using continuous values to approximate discrete data. The generator has 3 layers (256, 512, 3079), and the discriminator also has 4 layers (3079, 512, 128, 1). 2)T-wGAN: The generator is the same as wGAN. The encoder has 2 layers (512,64). The discriminator has 3 layers (64,32,1). 3)medGAN: The generator uses a 4-layer (256, 256, 256, 300) and adopts pre-training with autoencoder. The discriminator structure is the same as T-wGAN. 4)ehrGAN: Since our data has no temporal information, we just adopt the general framework of ehrGAN, without the structure of *word2vec* and convolutions. We use the same feature construction approach as medGAN, and uses a full connection network instead of convolution. The discriminator structure is the same as T-wGAN. 5)CorrGAN: In order to show the effectiveness of Grouped CorrNN, we also employ CorrNN in the framework of GAN. Specifically, for a fair comparison, we combine CorrNN and T-wGAN, and the main settings are the same as GcGAN_{mlp}. 6)GcGAN_{mlp}: The generator uses a 3-layer (256, 512, 300) fully connected network. The discriminator structure is the same as T-wGAN. 7)GcGAN_{den}: The generator has 4

layers (256, 256, 256, 300). The first three layers are connected with Dense connection, and the discriminator structure is the same as T-wGAN.

C. Evaluation Measures for Multi-label Classification

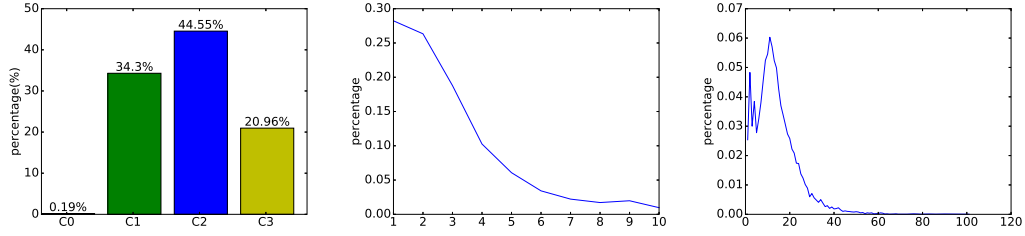
In order to measure the validity of the generated data, we use the multiple disease information of a patient to recommend the corresponding prescription, which can be viewed as a multi-label classification task (treatment recommendation). The performance of treatment recommendation will be used as an indicator in comparison of the quality of synthetic data. We demonstrate that using synthetic data can boost performances of predictive models (even for a simple model) just like the real-world data does. Hence we chose the most common multi-label learning algorithms instead of sophisticated ones, based on which the results will be more valuable in a general application. Two classical algorithm, i.e. RANdom k-labelsets (RAKEL) [18] and Multi-Label k-Nearest Neighbors (MLKNN) are used. Three popular metrics for evaluating multi-label classification, i.e. *Jaccard Similarity score*, *F1-score(micro)* and *Hamming Loss*, are used to evaluate the performances of the two algorithms with generated data using different models. The detailed information of the evaluation algorithms and metrics can refer to [18]. Note that the larger the value of Jaccard score and F1-score, the better, while the smaller the value of Hamming Loss, the better.

D. Results and Discussion

1) *Proportion of qualified synthetic data*: A qualified synthetic record should contain both diseases and drugs. If the synthetic record contains only diseases, or only drugs, it indicates that the model fails to effectively discover the relationship between the diseases and the drugs. We perform each model 10 times and generate 60,000 synthetic records each time. The mean and standard deviation of the proportion of qualified synthetic records are displayed in Table I, which shows that the GcGAN model has achieved excellent performance, and GcGAN_{den} has achieved the best results. In addition, the results demonstrate that learning a hidden layer representation (in medGAN, ehrGAN and our methods) is more effective than a direct learning of discrete variables (in wGAN and T-wGAN).

2) *Frequency of data features*: We investigate whether the generated data from each models are able to simulate original data, by comparing the frequencies of top 50 features of the real data and synthetic data. Fig. 4 shows that GcGAN_{den} keeps the most similar frequencies with the real data, and GcGAN_{mlp} ranks the second. We further calculate the feature co-occurrences to inspect whether GcGANs well learn the correlation between features. We calculate the difference value of co-occurrence frequency of top 50 features using different GANs, which can compare the consistency level more clearly. It can be calculated as follows

$$\sum_{i=0}^k \sum_{j=0}^k (p_{ij} - \hat{p}_{ij})^2$$



(a) The distribution of efficacy in the (b) The distribution of the number of (c) The distribution of the number of
records. diseases in each record drugs in each record.

Fig. 3. (a): C1, C2, and C3 represent “Cured”, “Improved”, and “Effective”, respectively. C0 represents the number of missing values. In experiment, we fill in the missing values with C1, C2, and C3 according to their proportion after removing C0. (b) (c): the x-axis represents the number of diseases and drugs in each record, respectively. The y-axis denotes the proportion to all records.

TABLE I
PROPORTION OF QUALIFIED SYNTHETIC DATA

model	wGAN	T-wGAN	medGAN	ehrGAN	CorrGAN	GcGAN _{mlp}	GcGAN _{den}
mean	0.8614	0.8762	0.8989	0.9328	0.9411	0.9512	0.9521
std.	0.0015	0.0013	0.0013	0.0008	0.0012	0.0012	0.0010

TABLE II
THE DIFFERENCE VALUE OF CO-OCCURRENCE FREQUENCY OF TOP 20 FEATURES

model	wGAN	T-wGAN	medGAN	ehrGAN	CorrGAN	GcGAN _{mlp}	GcGAN _{den}
mean	0.40620	0.34143	0.23130	0.12621	0.12522	0.10543	0.08987
std.	0.01056	0.01150	0.01134	0.00834	0.00723	0.00661	0.00630

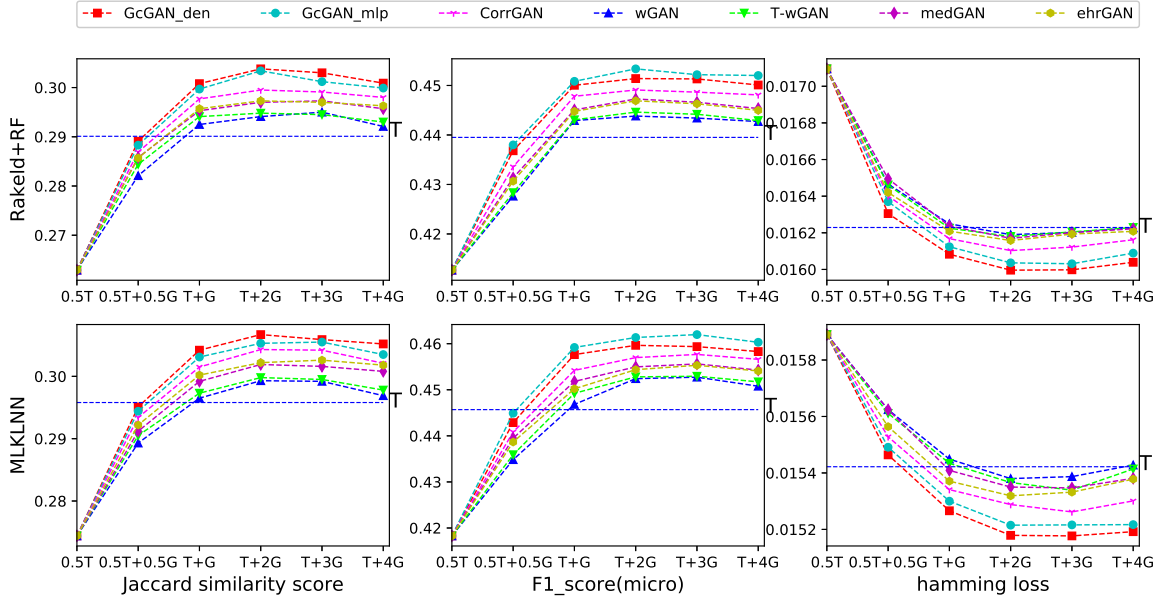


Fig. 5. The blue dotted line in the figure indicates the results when the training data is T. The first row denotes the results of RAKEL, while the second one denotes MLKNN. The first column, the second column, and the third column are Jaccard Similarity score, F1-score (micro), and Hamming Loss, respectively.

where p_{ij} , $\hat{p}_{ij}$ represent the co-occurrence frequency of the i -th feature and the j -th feature in real data and generated data, respectively. In the experiment, k is set to 50. We perform each model 10 times and generate 60,000 synthetic records each time. The mean and standard deviation of the difference value are displayed in Table II. Note that the results in Table II are

consistent with Table I. It shows that ehrGAN perform much better than wGAN, T-wGAN and medGAN, while GANs using correlation information perform the best. The fact that GcGANs outperform CorrGAN demonstrates the superiority of grouped CorrNN over CorrNN.

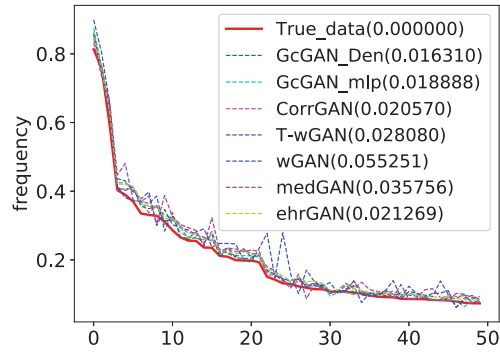


Fig. 4. The x-axis indicates the feature, while the y-axis indicates the frequency. The values in bracket depict the difference between the generated data and the real data, using the sum of the squares of the frequency differences.

3) *Performance in treatment recommendation:* RAKEL and MLKNN are used in treatment recommendation. For RAKEL, random forest is used as the base classifier. The number of label subsets and the number of RF evaluators are fine-tuned in the experiments, while other parameters are set to default values. For MLKNN, the parameter k in MLKNN is tuned. Since it is not the focus of this work, the effect of parameters of RAKEL and MLKNN are not reported in this paper.

We record the performances of multi-label classification with different size of training data as well as synthetic data generated using different approaches. We divide the data into training set (80%), validation set (10%) and test set (10%). The role of the training set is two-fold. First, it is used as input data of the generative models to generate synthetic EHR data. Second, it is used together with the synthetic data to train the multi-label learning algorithms. The validation set is used for parameter tuning of RAKEL and MLKNN. Specifically, the results on the test set using RAKEL and MLKNN trained with half of the training data (denoted as $0.5T$) as well as all the training data (denoted as T) are obtained as baselines for comparison. For each generated model, different sizes of synthetic data are generated and then combined with the original training data. We denote the new training sets as $0.5T + 0.5G$, $T + G$, $T + 2G$, $T + 3G$, $T + 4G$ respectively, according to their composition. For example, $0.5T + 0.5G$ consists of half of the instances in training set T and the same size of generated data, while $T + G$ consists of the real training set T and the same amount of generated data. Multi-label classification models are then trained with these new training sets. We repeat the experiment for 10 times and report the mean of the evaluation metrics of each model.

Fig. 5 shows that the results are consistent using the two algorithms with respect to all the three metrics. Generally, the synthetic data can boost the performance of multi-label classification. Compared to the results on $0.5T$, all the generated approaches are able to improve the classification performances by generating and adding the synthetic data $0.5G$ into the

training set. In this case, records generated using our models even achieve similar effects to the training set T (denoted by the blue dotted line in Fig. 5). It is natural that the results of $0.5T + 0.5G$ are slightly worse than the real data T since the quality of synthetic data may not be as good as the real data. As we continue, i.e. using the sets $T + G$, $T + 2G$, $T + 3G$, $T + 4G$ respectively, all the approaches are able to boost the performances continuously, gradually approaching their respective upper limits. Note that the metrics of all approaches even deteriorate when the training set is $T + 4G$. The main reason may lie in the fact that more noisy or inconsistent information in the synthetic data are introduced to the multi-label classification models when more synthetic data are added. Among the methods, our models outperform other approaches significantly, while CorrGAN ranks the 3rd. It demonstrates again that considering the correlation between variables is indeed beneficial to the GAN model. Compared with CorrNN, the proposed Grouped CorrNN captures the grouping information of drugs, so it further improves the performance of GANs. Fig. 5 also shows that the medGAN and ehrGAN achieve similar performances with each other, and outperform T-wGAN and wGAN.

A generated record

ICD codes: P39.200, P24.001
Drug codes: 300024, 300033, 300338, 300372, 300878, 300895, 301719, 301990, 303651, 314051

Meaning of the code

Code	Corresponding Disease or Drug
P39.200	Fetal amniocentesis
P24.001	Neonatal meconium aspiration syndrome
300024	Cefazolin sodium for injection
300033	Cefuroxime sodium for injection
300338	Heparin sodium injection
300372	Vitamin K1 injection
300878	10% glucose injection
300895	10% sodium chloride injection
301719	Zinc oil
301990	Sodium bicarbonate injection
303651	Injection Cefmetazole
314051	0.9% sodium chloride injection

Fig. 6. A generated record from GcGAN. ICD codes indicates International Classification of Diseases. P39.200, P24.001 are fetal amniocentesis and neonatal meconium aspiration syndrome. The drug codes in red are considered to be disease-specific for P39.200, P24.001 by GcGAN. Others indicate adjuvant drugs.

4) *The ability to distinguish prescription:* GcGAN is able to distinguish between disease-specific drugs and adjuvant drugs in the treatment, which make it more interpretable. In all the generated data, we decode h_{i1} , h_{i2} , h_{i3} by D2 to automatically distinguish disease, disease-specific drugs and adjuvant drugs, respectively. To show the result more intuitively, an example of generated data is shown in Fig. 6. The drug codes in red represent the disease-specific drugs decoded from h_{i2} , while others are adjuvant drugs. It can be observed that GcGAN can distinguish between these two groups of drugs.

Further, we examine the frequency of the generated drugs occurrence in the real data. Fig. 7 shows the frequency of two groups of drugs in real records. Note that the two groups of drugs come from the decoding process in GcGAN when we generate 10,000 synthetic records. It shows that the frequency of most drugs decoded as “disease-specific” drugs is much lower than that of “adjuvant drugs”. This result verifies the rationality of our approach because adjuvant drugs can occur in many treatment regimens while disease-specific drugs are only used in specific diseases. Part of this automatic classification have been verified by the doctors. Note that other GAN models may generate similar frequency distributions in Fig. 7, but they cannot automatically distinguish different groups of drugs in each generated record.

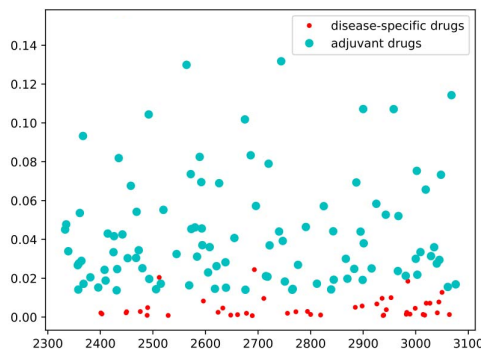


Fig. 7. The frequency of the drugs in the real records. The x-axis denotes the index (2332-3079) of the drug, and the y-axis denotes the frequency divided by the total number of records.

V. CONCLUSION AND FUTURE WORK

In this work, we propose GcGAN to generate discrete EHR data. In the model, we consider the structure of EHR data, and design Grouped CorrNN to better discover the correlation between disease and drugs and to distinguish disease-specific drugs and adjuvant drugs in the treatment. GcGAN has achieved impressive performance compared to state-of-the-art approaches, and verifies the benefits of using grouped correlation information to improve GANs. The idea can also be easily extended to datasets that contain correlated grouped of features, or other background knowledge of the grouping of variables. In the future, we will explore the possibility of using efficacy as reward and design a reinforcement learning based technique in order to further utilize the information.

REFERENCES

- [1] E. Choi, S. Biswal, B. Malin, J. Duke, W. F. Stewart and J. Sun. “Generating multi-label discrete patient records using generative adversarial networks.” *Machine learning for healthcare conference*. 2017.
- [2] P. Isola, J.-Y. Zhu, T. Zhou, and A. A. Efros. “Image-to-image translation with conditional adversarial networks.” *Proceedings of the IEEE conference on computer vision and pattern recognition*. 2017.
- [3] I. Goodfellow, J. Pouget-Abadie, M. Mirza, B. Xu, Bing, D. Warde-Farley, S. Ozair, A. Courville, and Y. Bengio. “Generative adversarial nets.” *Advances in neural information processing systems*. 2014.
- [4] A. Yahi, R. Vanguri, and N. Elhadad and N. P. Tatonetti. “Generative Adversarial Networks for Electronic Health Records: A Framework for Exploring and Evaluating Methods for Predicting Drug-Induced Laboratory Test Trajectories.” *arXiv preprint arXiv:1712.00164*
- [5] T. Mikolov, I. Sutskever, K. Chen, G. S. Corrado, and J. Dean. “Distributed representations of words and phrases and their compositionality.” *Advances in neural information processing systems*. 2013.
- [6] R. J. Williams. “Simple statistical gradient-following algorithms for connectionist reinforcement learning.” *Machine learning* 8.3-4 (1992): 229-256.
- [7] C. Villani. “Optimal transport: old and new.” Vol. 338. Springer Science & Business Media, 2008.
- [8] C. Vondrick, H. Pirsiavash, and A. Torralba. “Generating videos with scene dynamics.” *Advances In Neural Information Processing Systems*. 2016.
- [9] S. Reed, Z. Akata, X. Yan, L. Logeswaran, B. Schiele, and H. Lee. “Generative adversarial text to image synthesis.” *International Conference on Machine Learning*. 2016.
- [10] Y. LeCun B. Boser, J. S. Denker, D. Henderson, R. E. Howard, W. Hubbard, and L. D. Jackel. “Backpropagation applied to handwritten zip code recognition.” *Neural computation* 1.4 (1989): 541-551.
- [11] L. Yu, W. Zhang, J. Wang, and Y. Yu. “Seqgan: Sequence generative adversarial nets with policy gradient.” *Thirty-First AAAI Conference on Artificial Intelligence*. 2017.
- [12] H. Zhang, T. Xu, H. Li, S. Zhang, X. Wang, X. Huang, and D. N. Metaxas. “Stackgan: Text to photo-realistic image synthesis with stacked generative adversarial networks.” *Proceedings of the IEEE International Conference on Computer Vision*. 2017.
- [13] A. Jolicoeur-Martineau. “The relativistic discriminator: a key element missing from standard GAN.” *arXiv preprint arXiv:1807.00734* (2018).
- [14] M. Arjovsky, S. Chintala and L. Bottou. “Wasserstein generative adversarial networks.” *International conference on machine learning*. 2017.
- [15] A. Odena, C. Olah, and J. Shlens. “Conditional image synthesis with auxiliary classifier gans.” *Proceedings of the 34th International Conference on Machine Learning-Volume 70*. JMLR. org, 2017.
- [16] W. N. Price, and I. Glenn Cohen. “Privacy in the age of medical big data.” *Nature medicine* 25.1 (2019): 37.
- [17] J. He, S. L. Baxter, J. Xu, J. Xu, X. Zhou, and K. Zhang. “The practical implementation of artificial intelligence technologies in medicine.” *Nature medicine* 25.1 (2019): 30.
- [18] G. Tsoumakas, I. Vlahavas. “Random k-labelsets: An ensemble method for multilabel classification.” *European conference on machine learning*. Springer, Berlin, Heidelberg, 2007.
- [19] G. Huang, Z. Liu, L. Van Der Maaten, and K. Q. Weinberger. “Densely connected convolutional networks.” *Proceedings of the IEEE conference on computer vision and pattern recognition*. 2017.
- [20] T. Miyato, T. Kataoka, M. Koyama, and Y. Yoshida. “Spectral normalization for generative adversarial networks.” *In ICLR*, 2018.
- [21] S. Chandar, M. M. Khapra, H. Larochelle, and B. Ravindran. “Correlational neural networks.” *Neural computation* 28.2 (2016): 257-285.
- [22] S. Ioffe, and C. Szegedy. “Batch normalization: Accelerating deep network training by reducing internal covariate shift.” *International Conference on Machine Learning*. 2015.
- [23] B. K. Beaulieu-Jones, Z. S. Wu, C. Williams, R. Lee, S. P. Bhavnani, J. B. Byrd, and C. S. Greene. “Privacy-preserving generative deep neural networks support clinical data sharing.” *Circulation: Cardiovascular Quality and Outcomes* 12.7 (2019): e005122.
- [24] S. L. Hyland, C. Esteban, and G. R’tsch. “Real-valued (medical) time series generation with recurrent conditional gans.” *stat* 1050 (2017): 8.
- [25] Z. Che, Y. Cheng, S. Zhai, Z. Sun, and Y. Liu. “Boosting deep learning risk prediction with generative adversarial networks for electronic health records.” *2017 IEEE International Conference on Data Mining (ICDM)*. IEEE, 2017.
- [26] J. Su. “Training Generative Adversarial Networks Via Turing Test.” *arXiv preprint arXiv:1810.10948* (2018).
- [27] E. Choi, M. T. Bahadori, J. Sun, J. Kulas, A. Schuetz, and W. Stewart. “Retain: An interpretable predictive model for healthcare using reverse time attention mechanism.” *Advances in Neural Information Processing Systems*. 2016.
- [28] S. Patel, A. Kakadiya, M. Mehta, R. Derasari, R. Patel, and R. Gandhi. “Correlated discrete data generation using adversarial training.” *arXiv preprint arXiv:1804.00925* (2018).
- [29] Y. Zhang, Z. Gan, and L. Carin. “Generating text via adversarial training.” *NIPS workshop on adversarial training*. Vol. 21. 2016.
- [30] Y. Choi, M. Choi, M. Kim, J. Ha, S. Kim, and J. Choo. “Stargan: Unified generative adversarial networks for multi-domain image-to-image translation.” *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*. 2018.